SharkVision

Exploring deep learning for shark species classification from BRUVS videos

Tedi Yankov, Ed Lucas, Pragya Singh

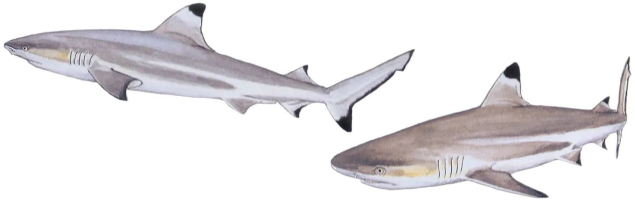
Contents

Introduction

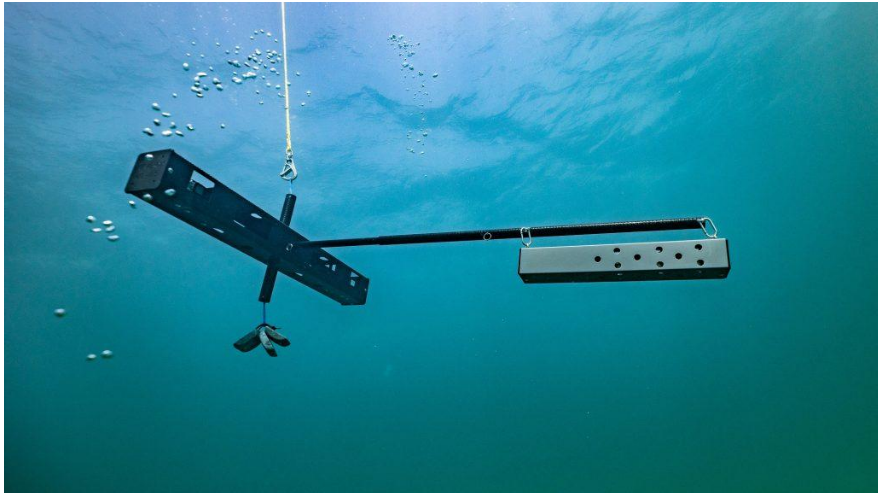
- ➔ Continuous and long-term ecosystem monitoring is an important process that can provide a warning of biodiversity loss and adverse environmental changes (Pasher et al., 2014).
- ➔ Scientists are tasked with *observing* without *interfering* → use remote sensing and monitor through images and videos (Marini et al. 2017)
- ➔ **Problem**: technological progress has enabled data generation at much higher frequencies over much larger areas, increasing the data volume to the point where manual labelling is not feasible (Oba and Doi 2025)
- ➔ **Solution**: computer vision with deep learning!

Data

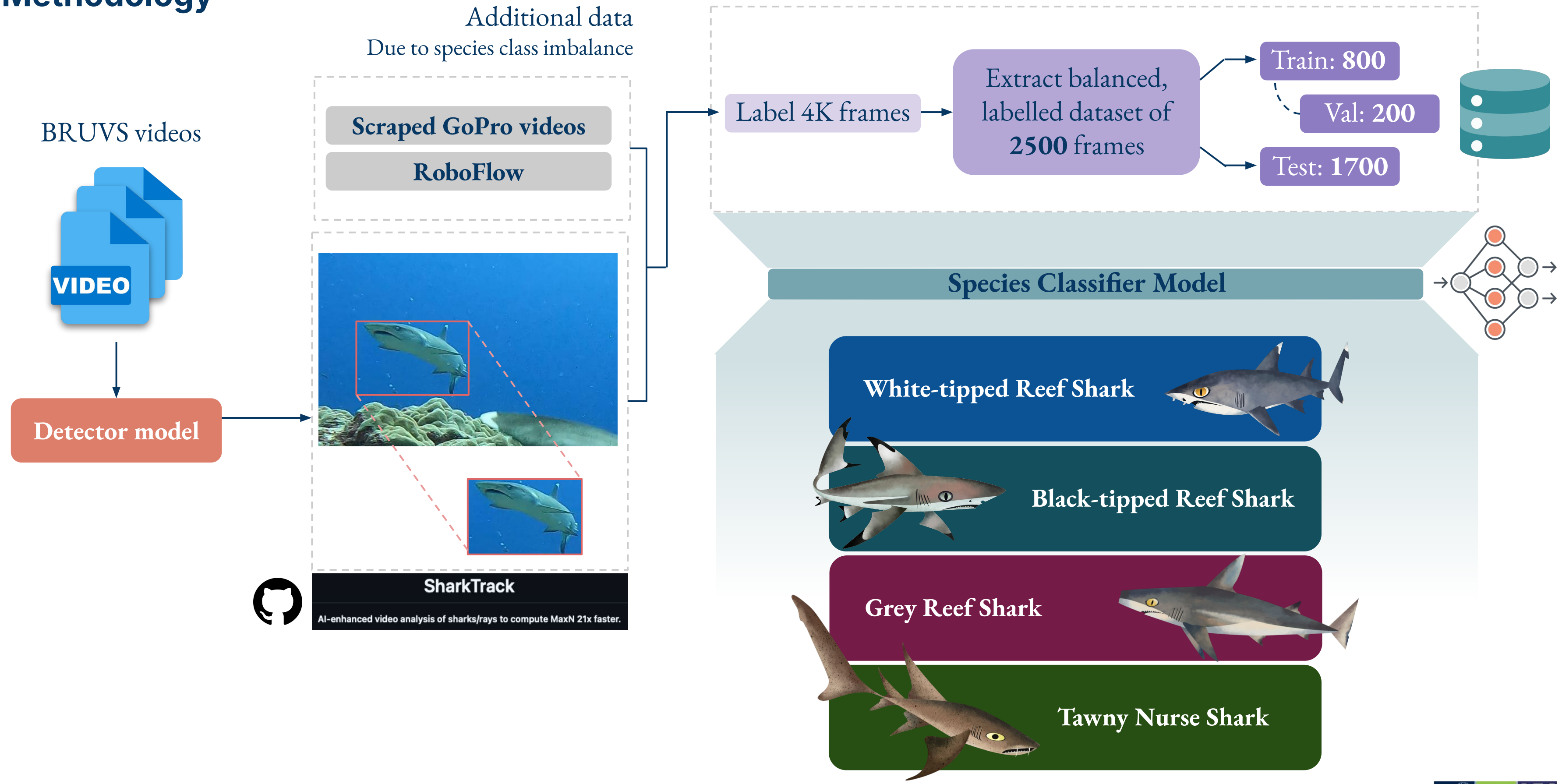
Baited Remote Underwater Video Systems (BRUVS)



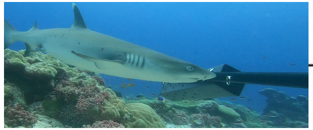
Can you spot the shark?

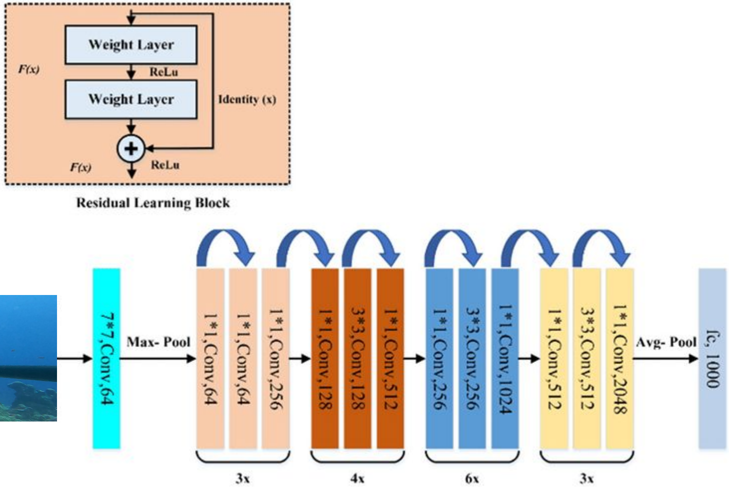


Methodology



Approach 1: ResNet50 baseline

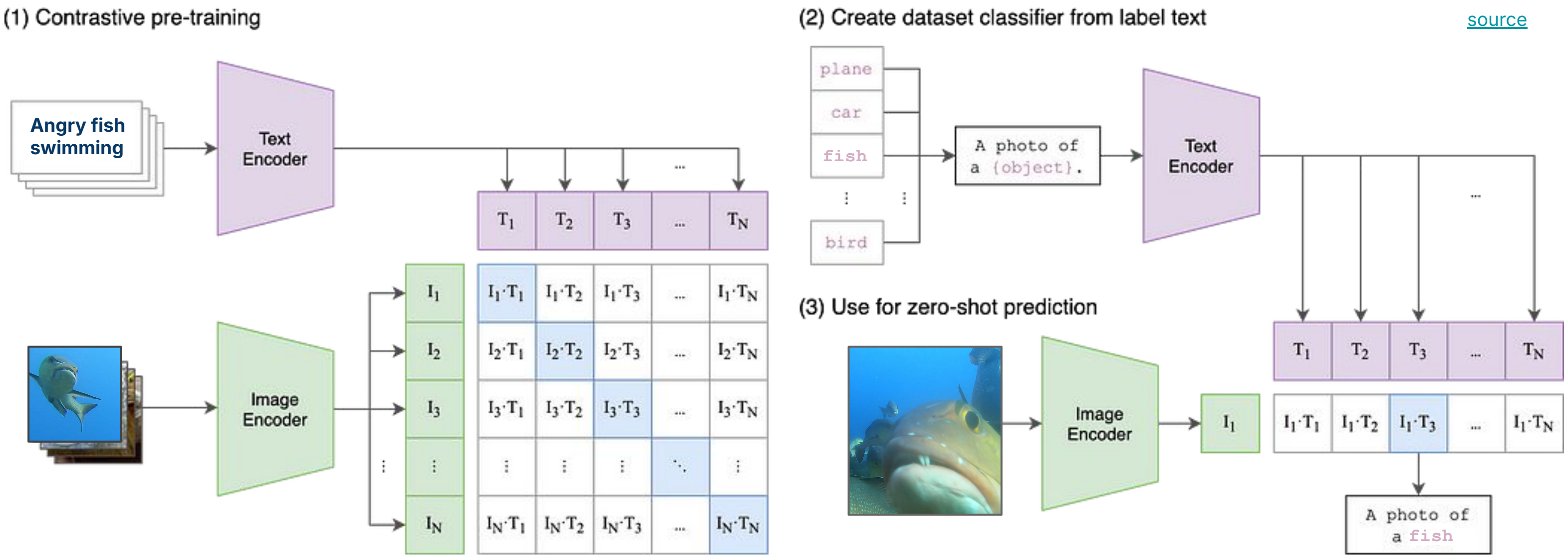




source

- ➔ Deep CNN with 50 layers, designed for image classification.
- ➔ Uses "residual connections" (skip connections) that allow gradients to flow directly through the network.
- ➔ Pre-trained on ImageNet (1.2M images, 1000 classes), fine-tuned on Roboflow shark dataset.
- ➔ Strength: CNN inductive biases: built-in assumptions about images (locality, translation invariance, hierarchical features) make it efficient at learning visual patterns from limited data
- ➔ Limitation: supervised learning requires exact class overlap between training and deployment.

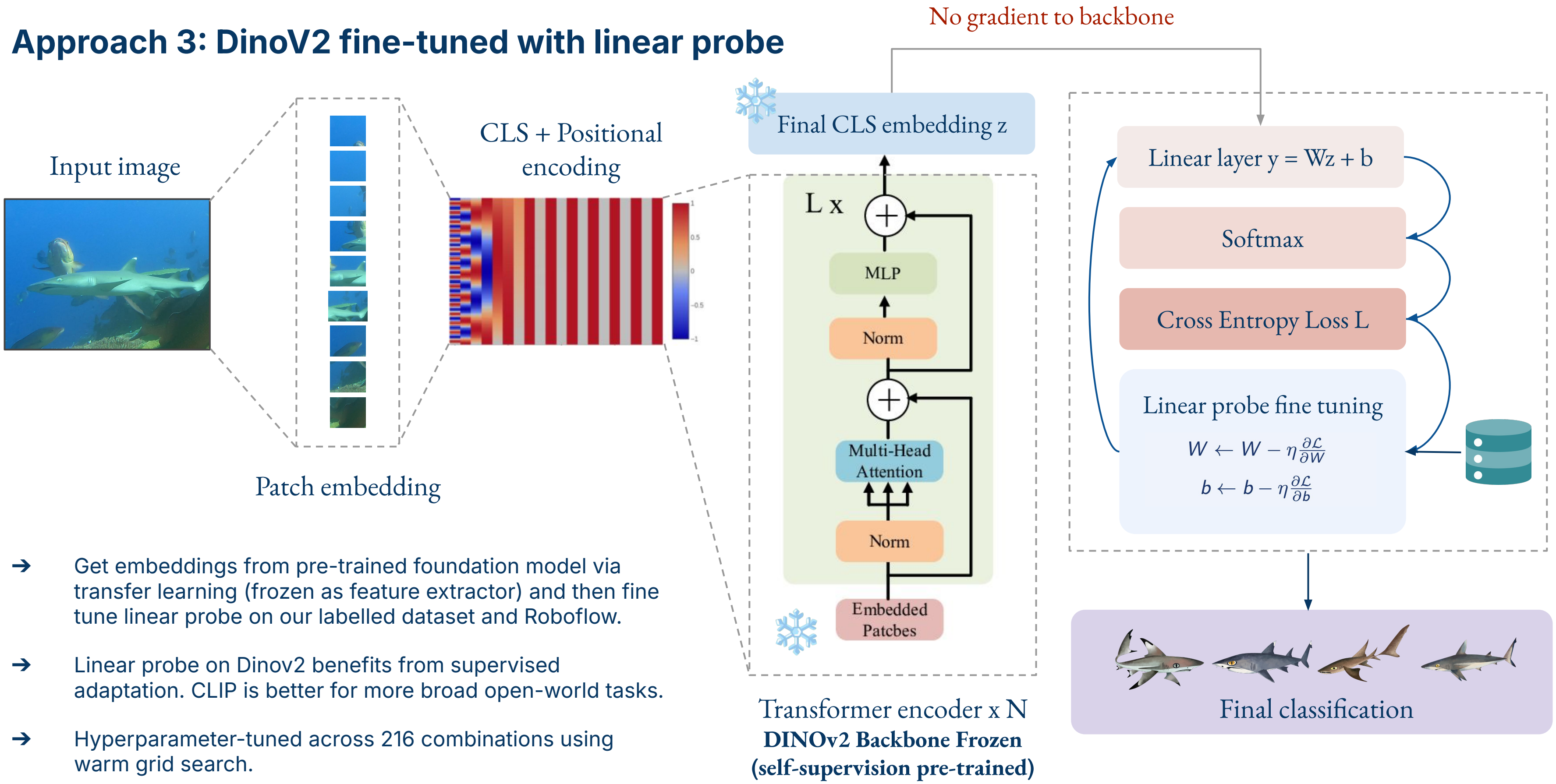
Approach 2: CLIP



source

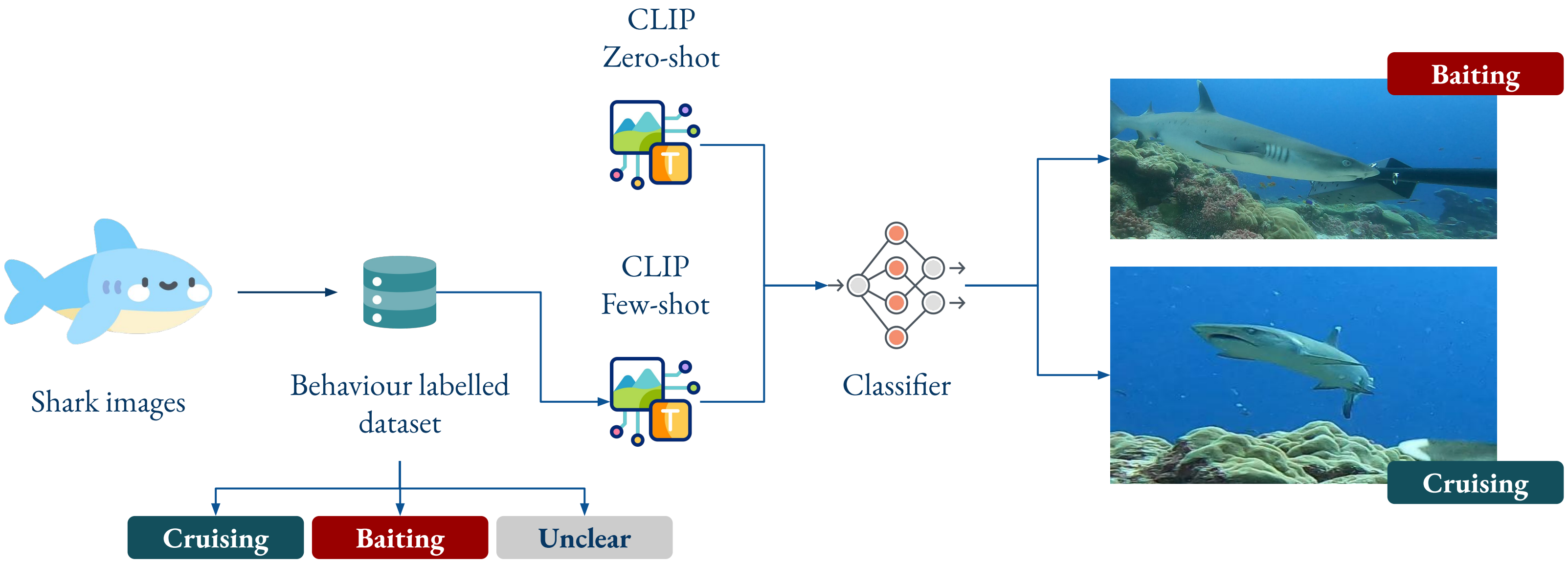
- ➔ Contrastive Language–Image Pretraining (CLIP) → learns to connect images and text by training on a many image-caption pairs, enabling it to classify images based on language descriptions without task-specific training.
- ➔ Good because it doesn't require labelled training data, accuracy may be limited for species that are visually very similar or poorly represented in its training data, especially for fine-grained distinctions.
- ➔ We implement two versions: zero-shot and few-shot.

Approach 3: DinoV2 fine-tuned with linear probe



- ➔ Get embeddings from pre-trained foundation model via transfer learning (frozen as feature extractor) and then fine tune linear probe on our labelled dataset and Roboflow.
- ➔ Linear probe on Dinov2 benefits from supervised adaptation. CLIP is better for more broad open-world tasks.
- ➔ Hyperparameter-tuned across 216 combinations using warm grid search.

Bonus task: behaviour classification



- ➔ Adapting the CLIP approach to classify behaviour instead of species.
- ➔ Exploit the fact that behaviour can be easily described with natural language → easier than having to label a lot of data.
- ➔ One issue with the bounding box overlap approach is depth perception, so this could be a viable alternative.

Results

Out of sample performance metrics across models

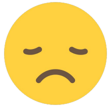
	Model	Accuracy	Precision	Recall	F1
Species classification	ResNet50 baseline	27.3%	69.8%	27.3%	34.0%
	CLIP zero-shot	50.7%	51.7%	50.9%	47.8%
	CLIP few-shot	68.8%	69.5%	68.9%	69.0%
	DINO (pre-tuning)	89.3%	90.1%	89.3%	89.5%
	⭐ DINO (post-tuning)	90.8% 98.9% top 3 accuracy	90.8%	90.8%	90.7%
Behaviour classification (using 1000-image manually labelled dataset)	CLIP zero-shot	33.6%			33.7%
	CLIP few-shot	61.9%			53.2%

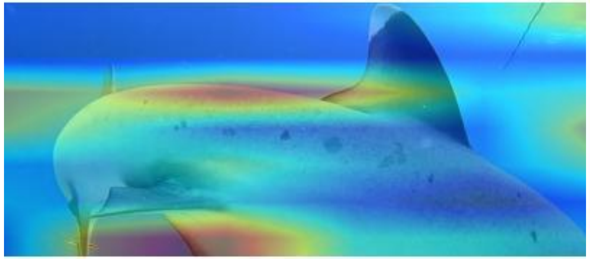
🎯Best DINO Hyperparameters:

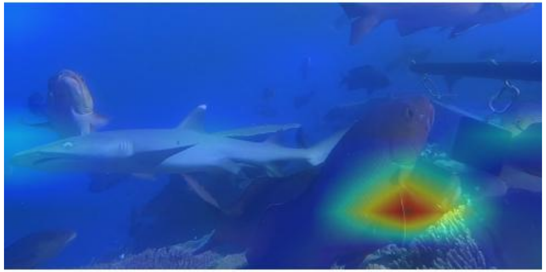
- learning_rate: 0.0001
- batch_size: 32
- hidden_dim: 256
- dropout: 0.3
- optimizer: adamw
- weight_decay: 0.0001

Results

Attention maps of example images, showing where the DINO model focuses its attention

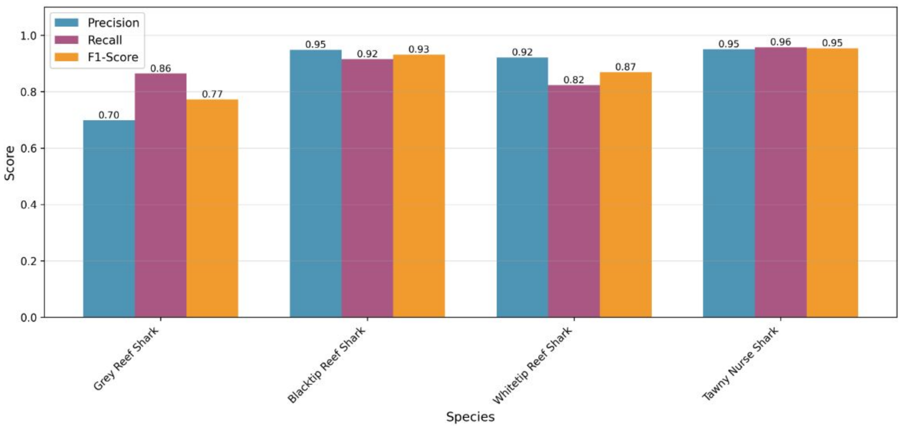




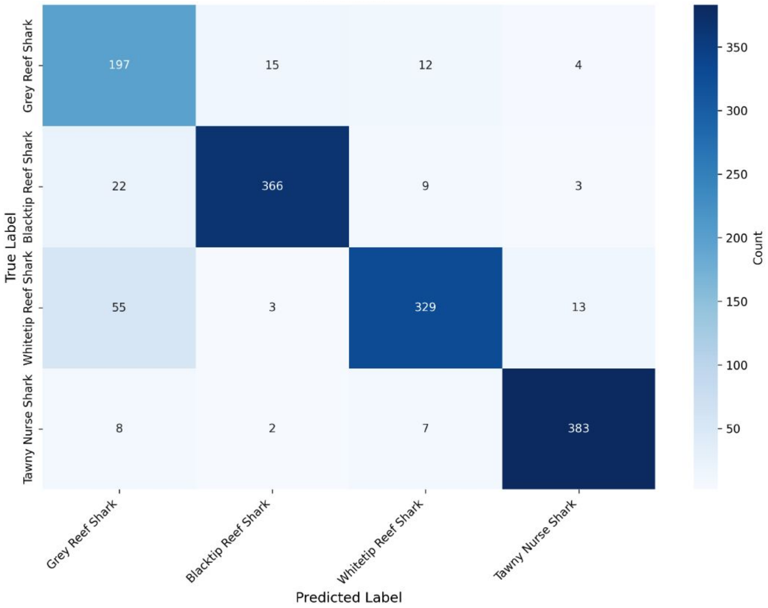


Results
Per-class metrics

Per-class performance



Confusion matrix



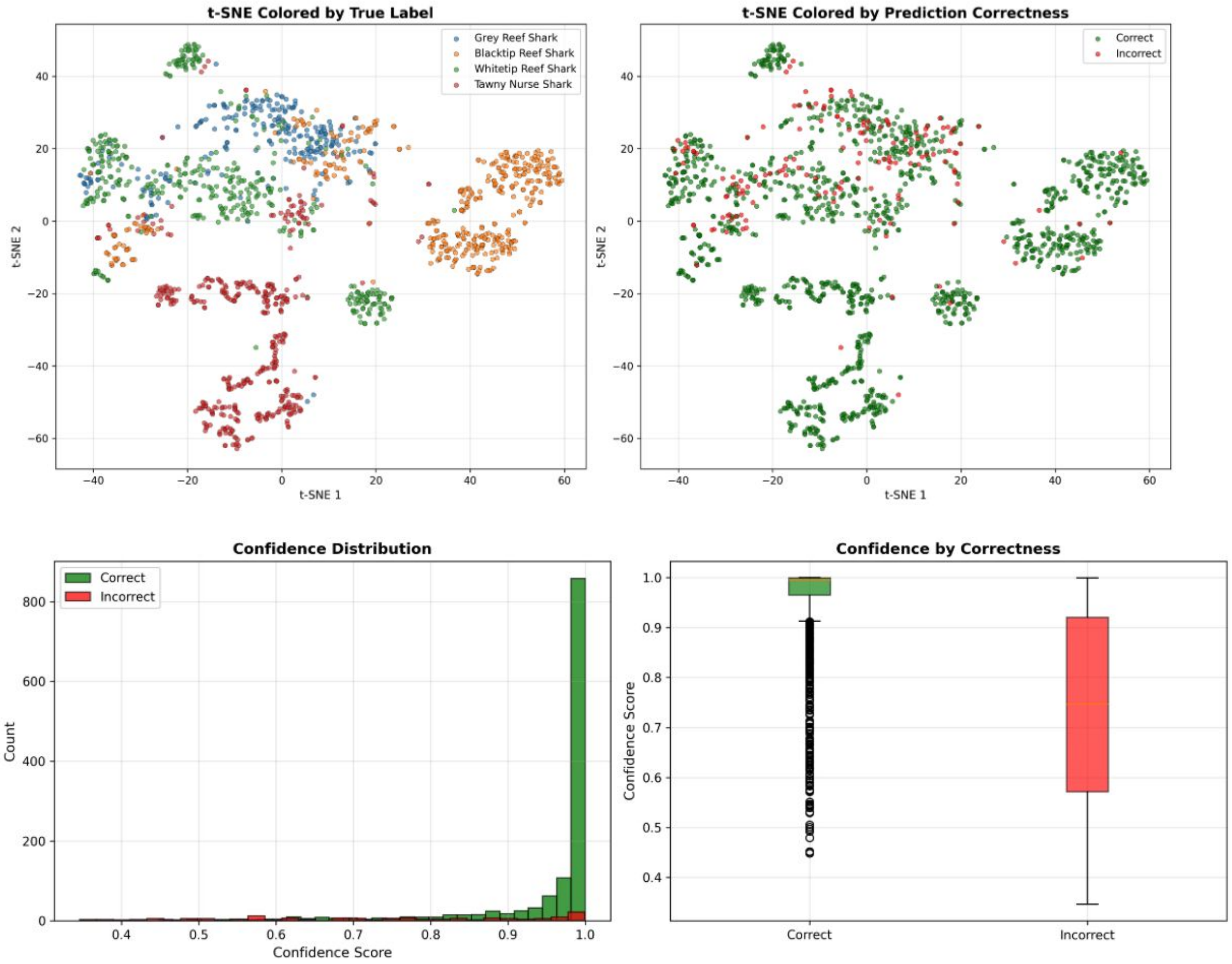
- 👍 High performance overall across classes, strong diagonal (means good species separation)
- 👎 Feature separation between reef species is weaker. Grey reef precision is relatively low.
- 🛠️ Partially unfreeze last transformer blocks, replace linear probe with small MLP, add more grey reef data.

Results
Feature quality

Learned features are meaningful and represent the true labels; correctness is helped by feature cluster separation; model is very confident when correct.

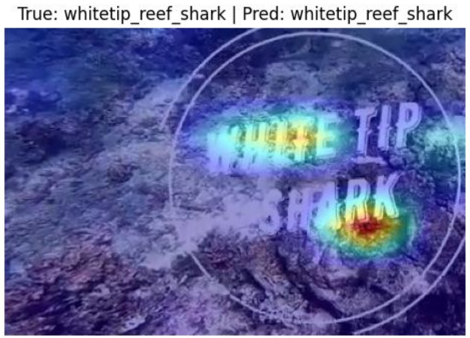
Model sometimes makes very confident mistakes... also is a bit more confused with reef species.

Partial fine tuning of backbone; small MLP; temperature scaling



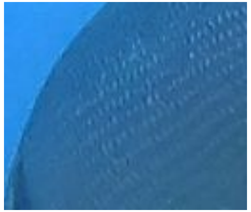
Limitations

➔ Need more/better data: class balance with BRUVS videos alone. Also, some of the images are... not ideal



➔ Gap filling with GoPro videos isn't necessarily the same as analysing more BRUVS videos.

➔ We rely on SharkTrack being good, but other detection approaches should be explored (eg SAM based models)



When the next shark to label is another white tip or grey reef

Thank you!

Questions?

Me looking for the dark matter shark that is allegedly in the single blue pixel image according to SharkTrack

The JASMIN ecosystem watching me submit a 16 hour hyperparam tuning job 3 seconds after getting orchid access